🛡️ Module 07: Perimeter Defence Evasion Techniques

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# Exercise 05: Using Proxychains to Avoid Detection and Bypass Perimeter Defences
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Scenario

Perimeter security infrastructure—such as firewalls, Intrusion Prevention Systems (IPS), and geolocation filters—frequently block scanning requests or direct connections originating from untrusted, unauthorized IP addresses. To conceal source attribution and bypass IP-based access filters, security testers route network traffic through intermediate proxies using **Proxychains**.

By configuring a local SOCKS proxy listener (such as Metasploit's **auxiliary/server/socks_proxy** bound to TCP port **9050**) and chaining traffic through **/etc/proxychains.conf**, scanning utilities like Nmap (**sudo proxychains nmap -sS 192.168.0.51**) transparently route requests through intermediate proxy hops. Scanning target machines (such as the FTP Server at **192.168.0.9**) via Proxychains reveals active listening services and identifies **6** open ports.

In this exercise, we review `/etc/proxychains.conf`, launch a SOCKS proxy server on port **9050** using Metasploit, verify port binding with `netstat`, execute Nmap scans through Proxychains against target IPs (**192.168.0.51** and **192.168.0.9**), and identify **6** open ports on the target.

Objective

The objective of this lab is to help students learn how to:
- Inspect and configure Proxychains settings in **/etc/proxychains.conf**
- Deploy Metasploit's SOCKS proxy auxiliary module (**auxiliary/server/socks_proxy**)
- Configure SOCKS proxy listening ports (**set SRVPORT 9050**)
- Verify active proxy listening ports using **sudo netstat -atn**
- Execute stealth SYN scans through SOCKS proxy chains (**sudo proxychains nmap -sS 192.168.0.51**)
- Determine open port counts (**6**) on target systems behind proxy chains

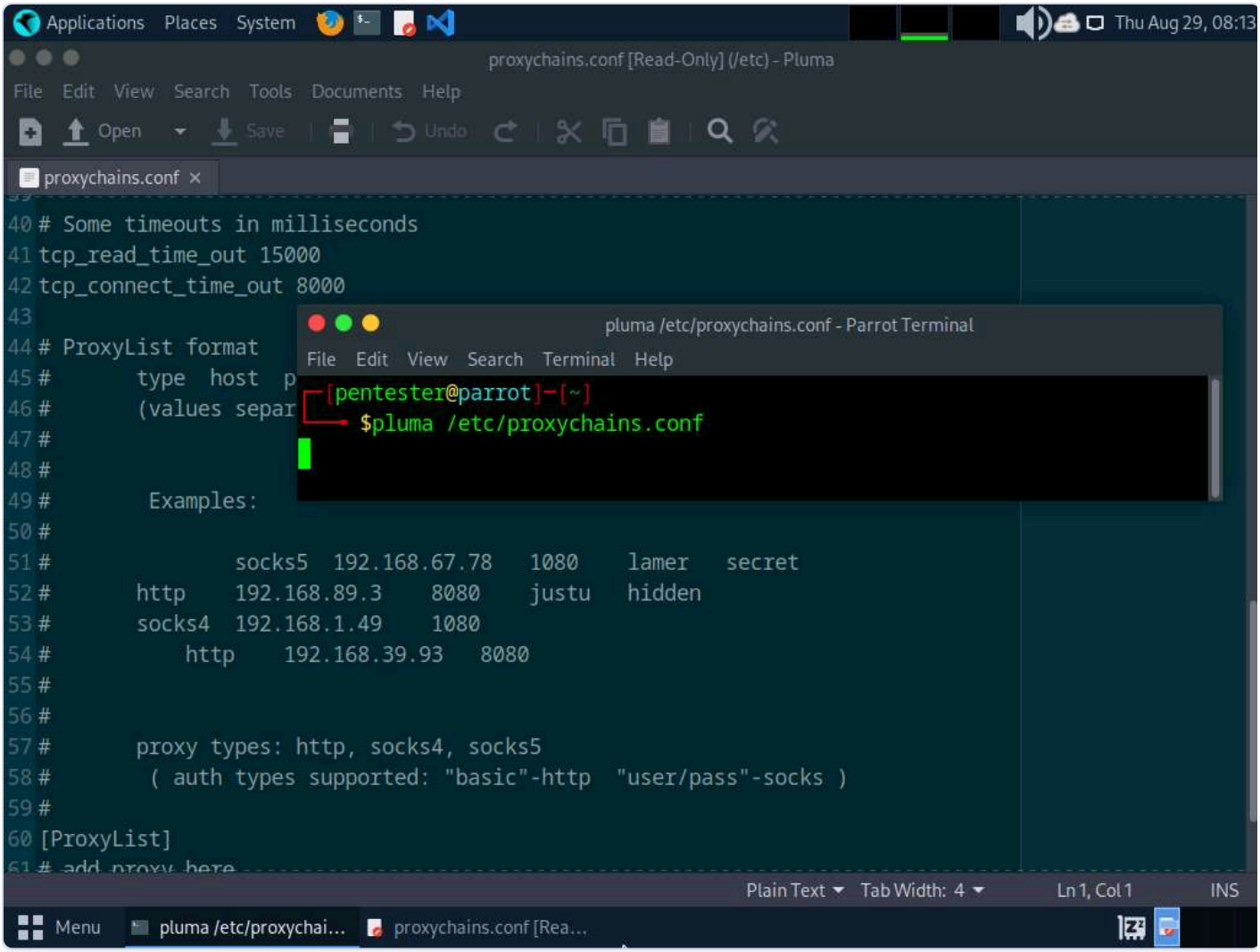
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Task 1: Configure SOCKS Proxy and Scan Targets via Proxychains

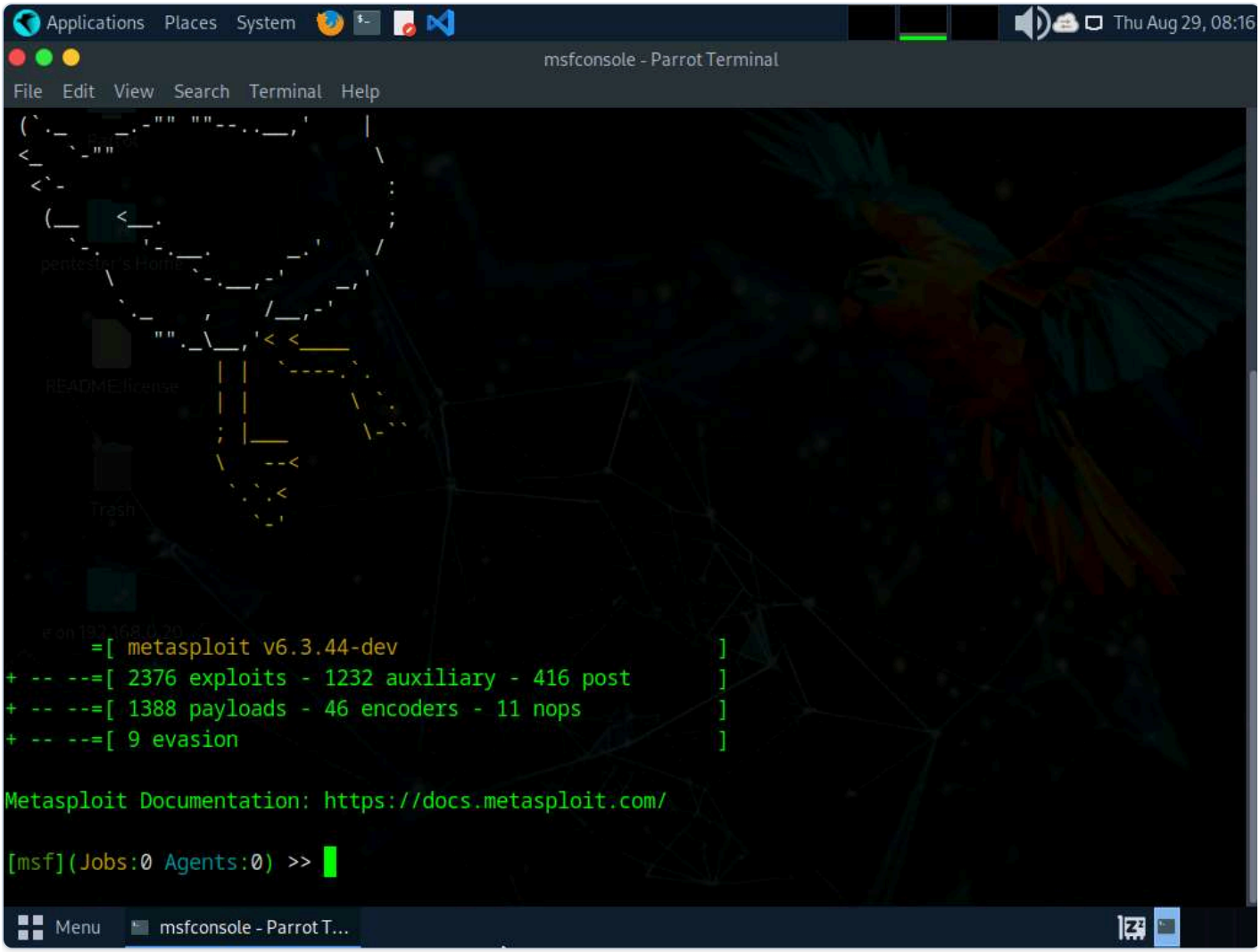
In this task, we review Proxychains configurations on Parrot Security, start a SOCKS proxy server on port 9050 in Metasploit, verify socket state, and run Nmap port scans through Proxychains.

Lab Tasks

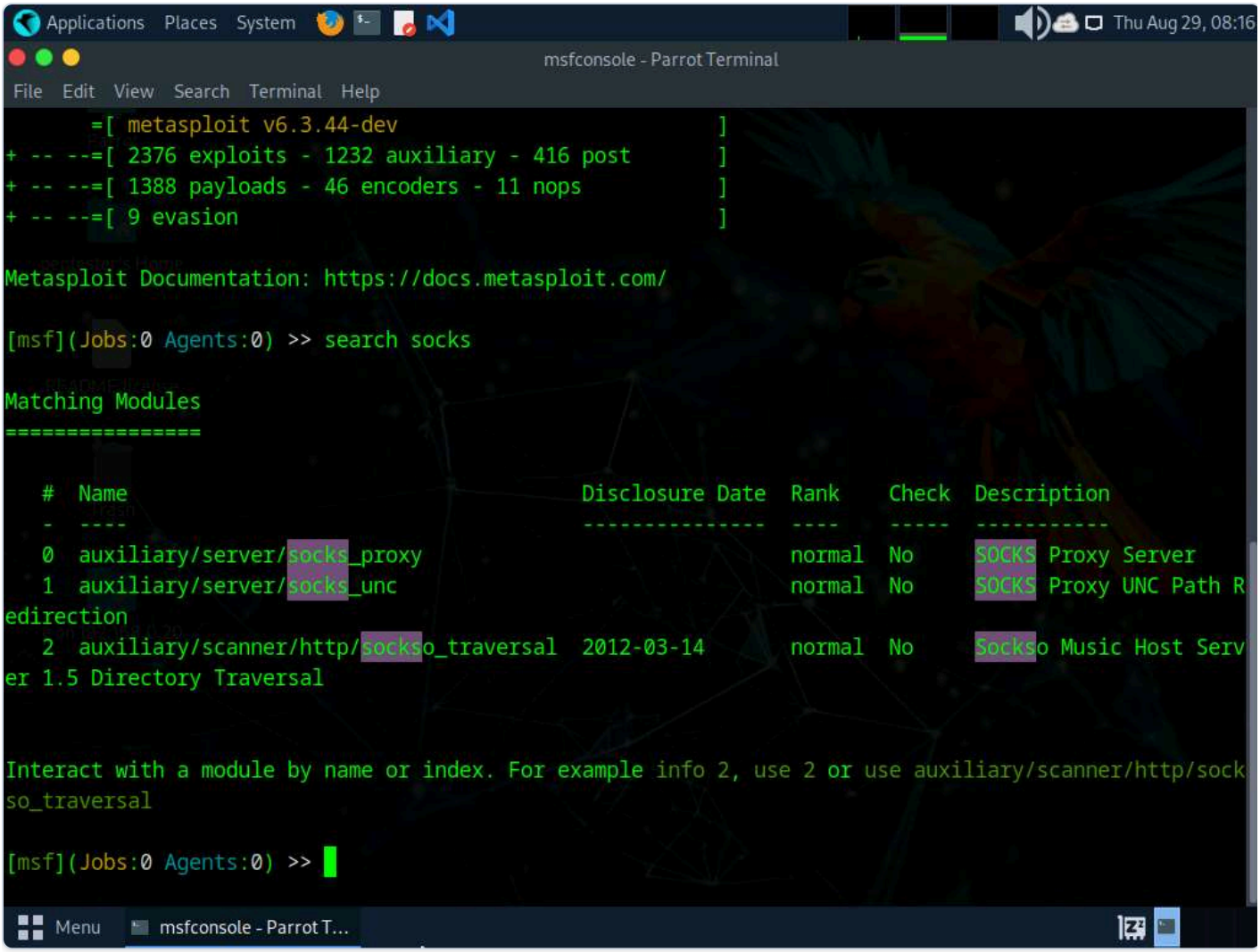
1. Switch to **Parrot Security** (**toor**). Open a terminal window and review Proxychains configuration:
   pluma /etc/proxychains.conf
   Inspect default proxy chain settings at the bottom of the file (`socks4 127.0.0.1 9050`), then close the text editor.



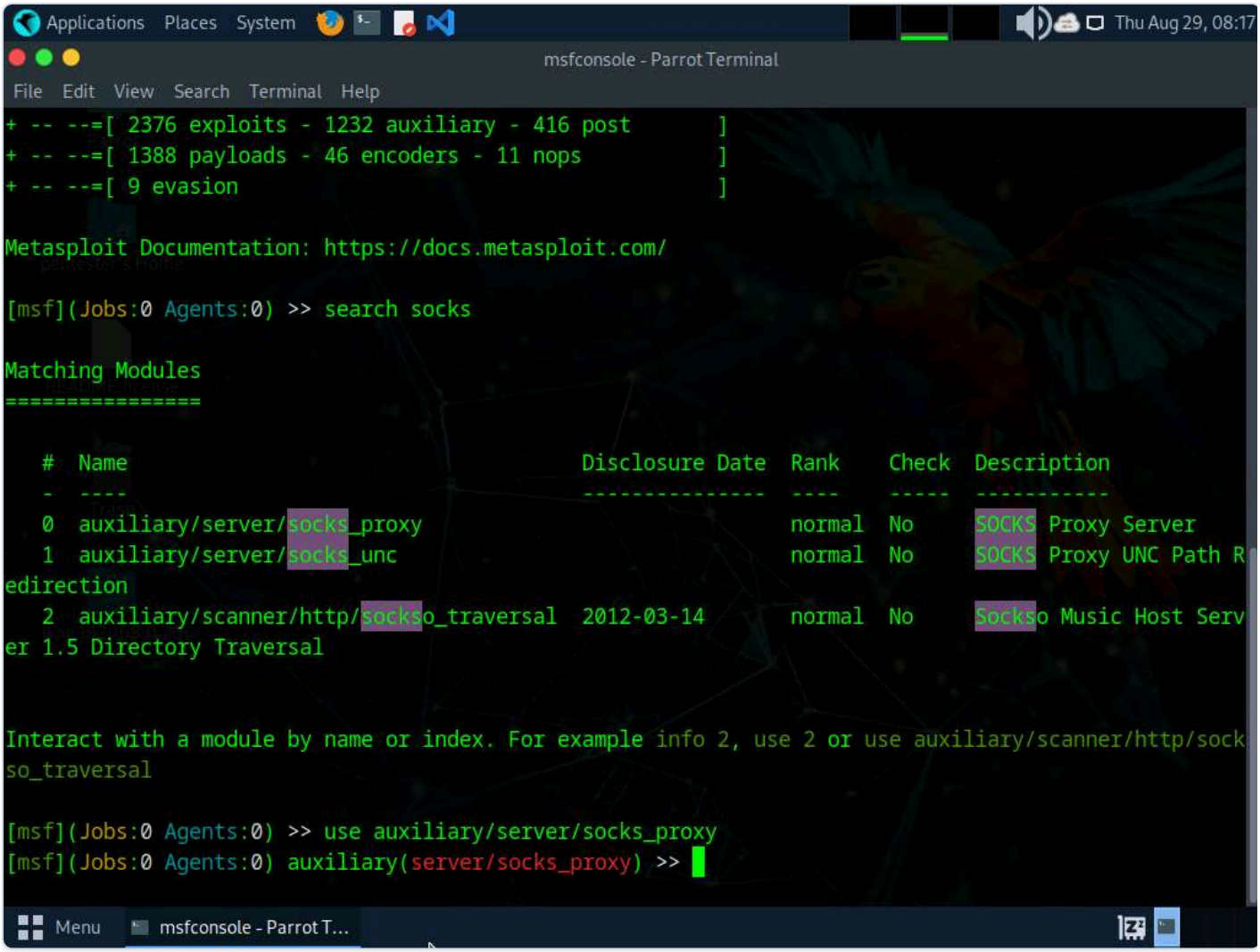
1. Launch Metasploit Framework:
   msfconsole



1. Search for available SOCKS proxy modules:
   search socks

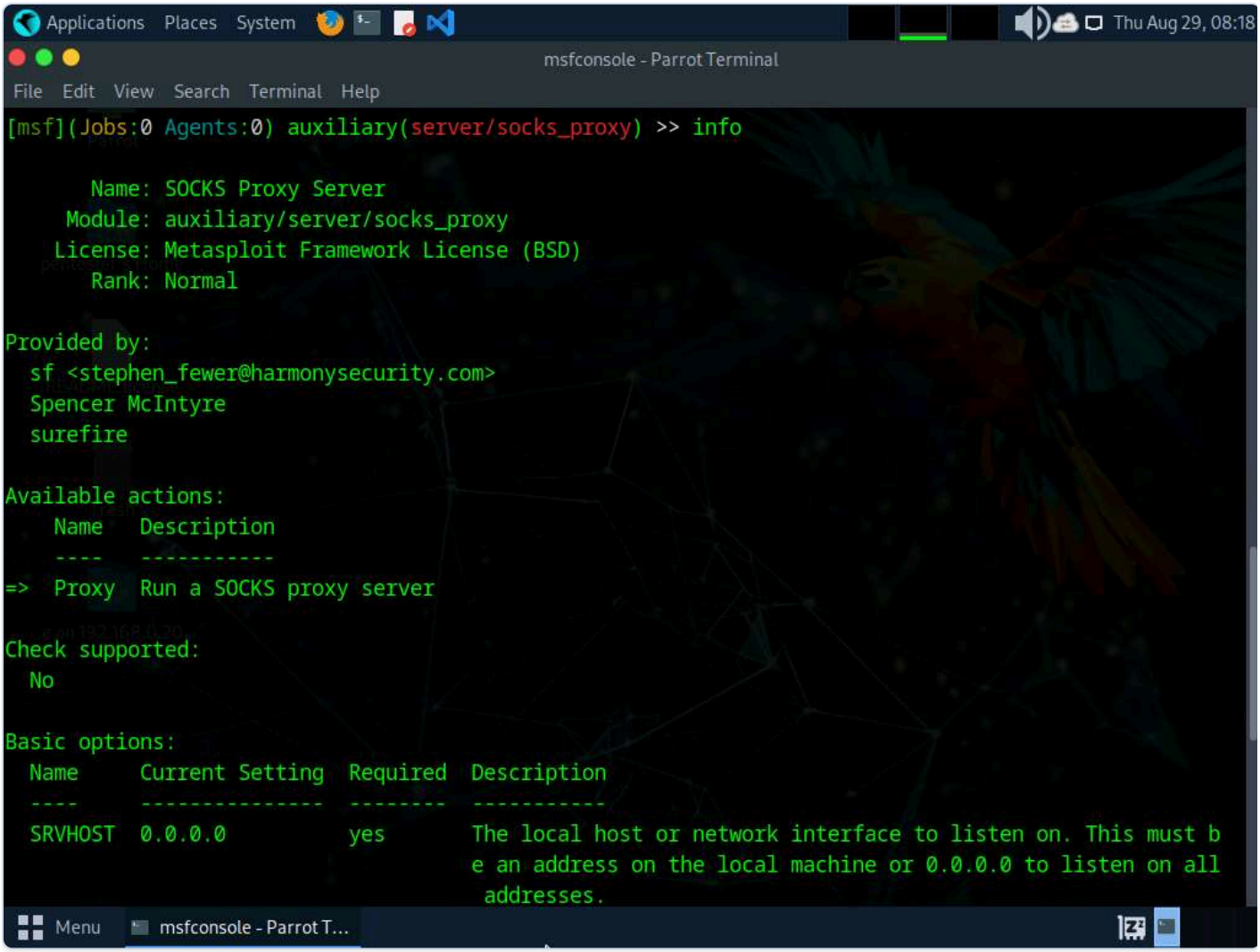


1. Load the SOCKS proxy server module:
   use auxiliary/server/socks_proxy

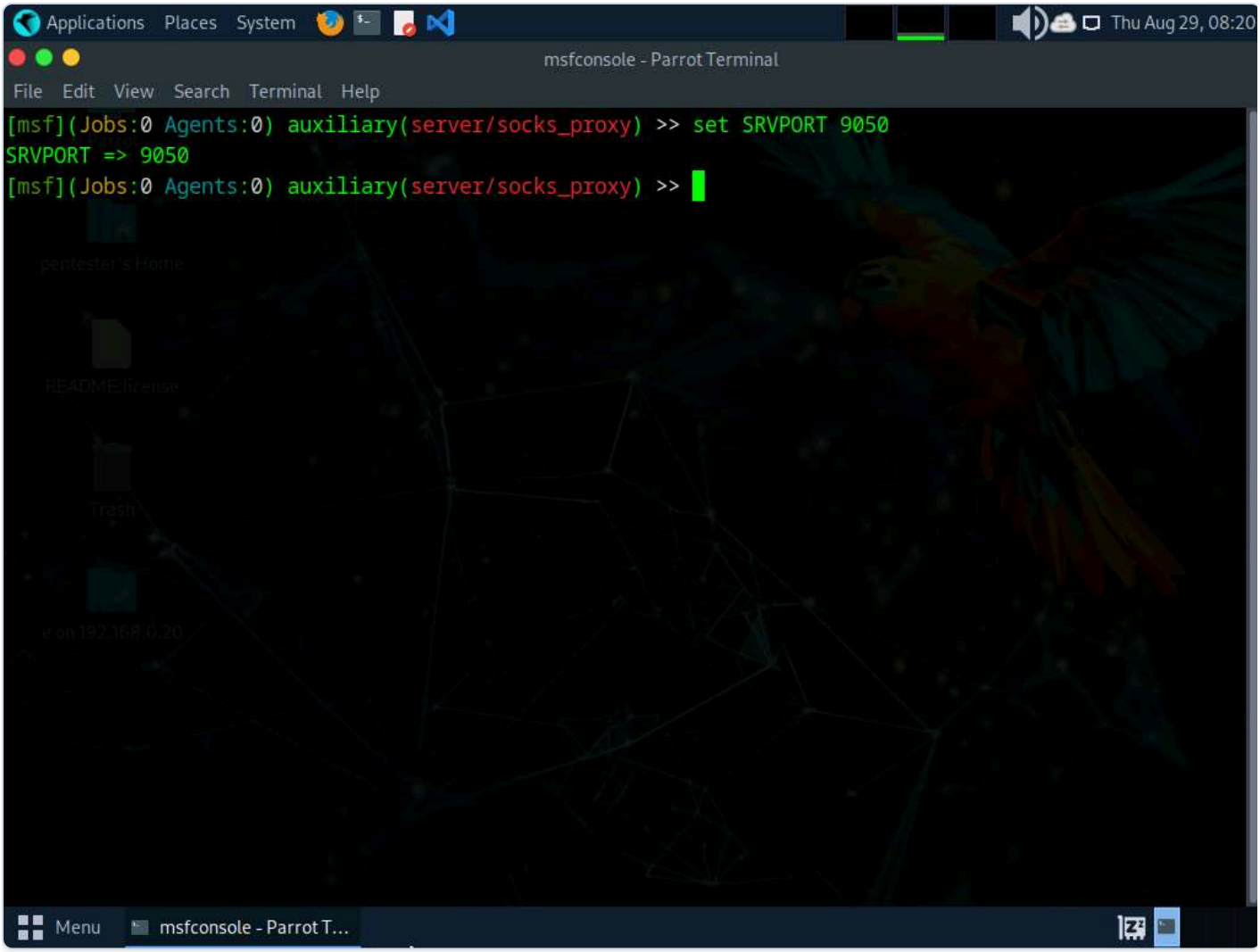


1. Inspect module options and descriptions:

   info

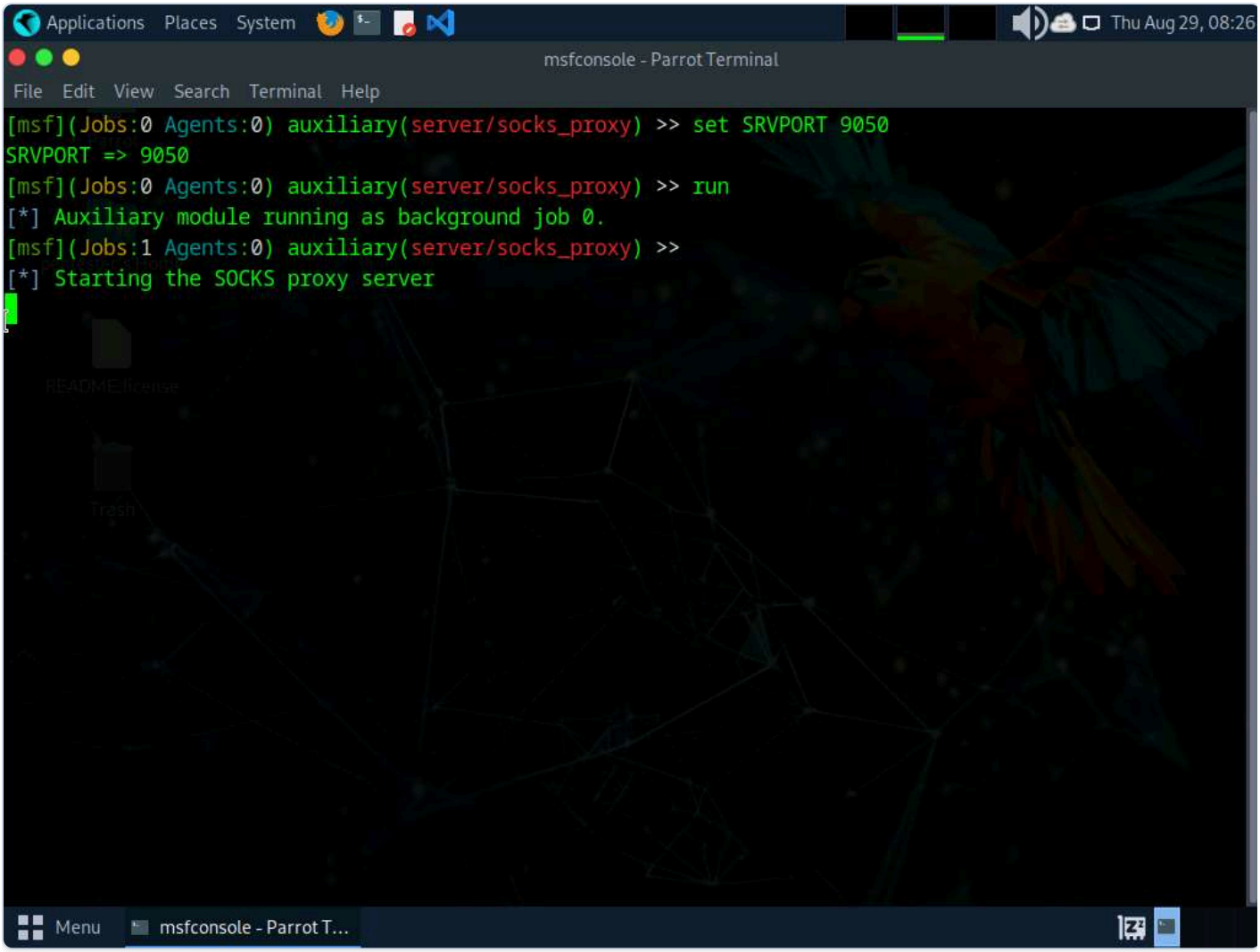


1. Set the server listening port to match Proxychains configuration:
   set SRVPORT 9050

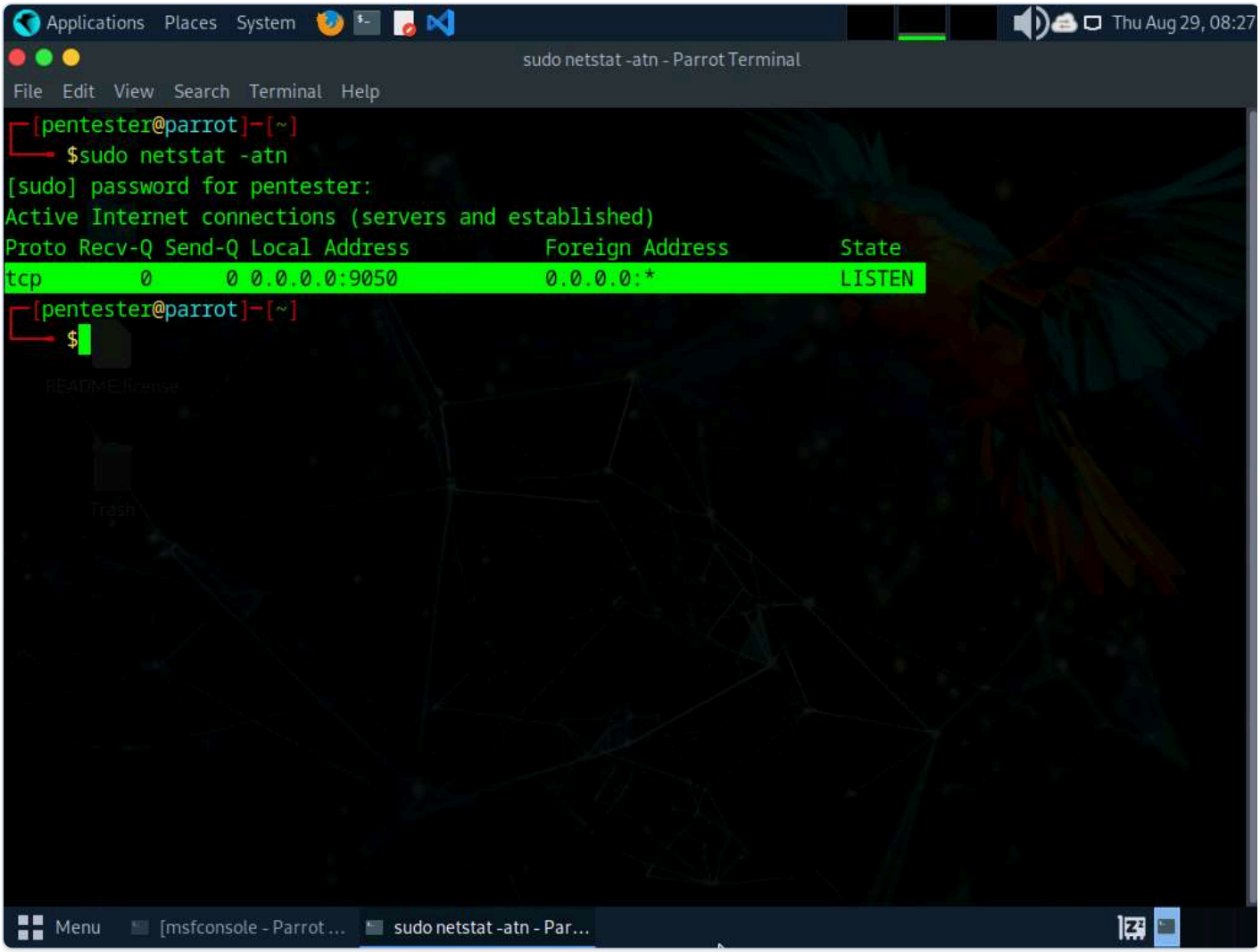


1. Start the SOCKS proxy server background job:

   run



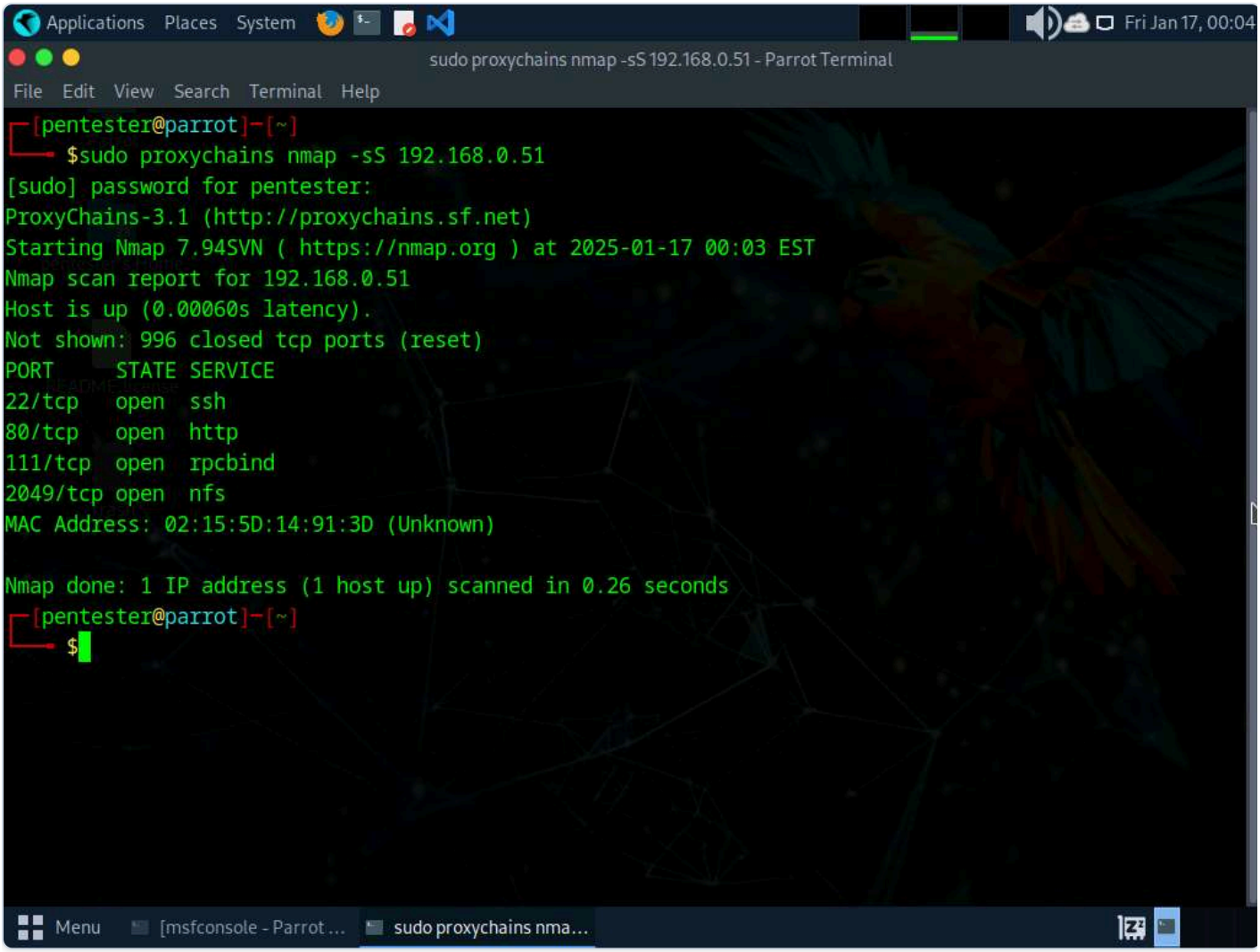
1. Open a new terminal tab/window and verify port 9050 listener binding:
   sudo netstat -atn
   (Confirm `0.0.0.0:9050` *is in* `LISTEN` *state).*



1. Execute Nmap SYN stealth scan routed through Proxychains against the target RPC Server (**192.168.0.51**) and FTP Server (**192.168.0.9**):

   sudo proxychains nmap -sS 192.168.0.51

   *Observe the tunneled SOCKS responses and verify that the target reveals **6** open ports.*



1. Close all open terminals and Metasploit sessions.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 7.5.1.1** On the Parrot Security machine, utilize ProxyChains to perform a port scan on the 192.168.0.50 (RPC Server - Ubuntu) to identify open ports. Additionally, use ProxyChains to scan the 192.168.0.9 (FTP Server) and determine the number of open ports. [!SUCCESS] ✅ **Answer:** **6**